



Practicum Manual

CYBER SECURITY OF POWER GRIDS

Course Code:
EE4410

Lecturer:
Dr. Alex Stefanov

Version 1.0
Updated: April 22, 2025

Table of Contents

Table of Contents	I
1 Lab Sessions and Assignments.....	3
1.1 Lab Sessions	3
2 URL Phishing, Malware and Passive Reconnaissance	4
2.1 Objectives	4
2.2 URL Phishing and Malware Identification.....	4
2.3 Passive Reconnaissance.....	6
2.4 Assignment	11
3 Cryptography Applications.....	13
3.1 Objectives	13
3.2 Introduction	13
3.3 Encryption and Decryption using AES	14
3.4 Encryption and Decryption using RSA	16
3.5 Elliptic Curves Digital Signature Algorithm (ECDSA)	17
3.6 Assignment	19
4 Cyber-Physical Data Correlation	20
4.1 Objectives	20
4.2 Anomaly Detection and Correlation of Power Grid Cyber-Physical Data.....	20
4.3 Assignment	20
4.4 References	20
5 Operational Technology Traffic Analysis.....	21
5.1 Objectives	21
5.2 Operational Technology Networks.....	21
5.3 Traffic Dispersion Graph.....	22
5.4 Assignment	23
5.5 References	23
6 Anomaly Detection and Time Series Classification	24
6.1 Objectives	24
6.2 Cyber-Physical Power System Co-Simulation	24

6.3	Anomaly Detection and Time Series Classification	26
6.4	Assignment	26
6.5	References	27
7	Report Guidelines	28
7.1	Report Requirements	28
7.2	Report Scoring Matrix	29
8	Access Guidelines	30

INTRODUCTION

Lab Sessions and Assignments

1.1 Lab Sessions

The course Cyber Security of Power Grids (EE4410) has five practicum modules divided across five lab sessions as given in Table 1.1. After the end of each lab session, students must do the practicum assignment(s) and submit the assignment report in time on Brightspace.

Assignments

- Each practicum has an assignment to be done in the lab and/or at home.
- Each student must write an individual assignment report for each practicum.
- The assignment reports (PDF) must be submitted on Brightspace, which will be checked automatically for plagiarism.
- The deadline for the report submission is one week after the lab session. Please check the report submission deadline on Brightspace.
- All assignments and reports are mandatory.

Table 1.1: Division of practicums per lab sessions

Lab Sessions	Activity	Platform	Assignment Reports
Lab 1	Practicum 1	Personal Computer	Max 5 PDF pages in total
Lab 2	Practicum 2	Personal Computer	Max 5 PDF pages and Jupyter Notebook
Lab 3	Practicum 3	Google Colab	Max 5 PDF pages and Jupyter Notebook
Lab 4	Practicum 4	Google Colab	Max 5 PDF pages and Jupyter Notebook
Lab 5	Practicum 5	Google Colab	Max 5 PDF pages and Jupyter Notebook

URL Phishing, Malware and Passive Reconnaissance

2.1 Objectives

- Explain URL phishing, malware, and passive reconnaissance.
- Demonstrate URL phishing and malware identification using publicly available tools.
- Demonstrate passive reconnaissance using the Open-Source Intelligence Tools (OSINT).

2.2 URL Phishing and Malware Identification

Phishing is defined as a social engineering cyber attack. This can be used to steal information and install malware on IT systems. One of the most common mechanisms to launch phishing is via email-based communication. Phishing can also be performed via website pages. Phishing attacks are often accompanied by a malicious Uniform Resource Locator (URL). The link will be directed to a malicious website and often contain malware. To prevent this phishing attack, a user must be aware of the malicious URL. Therefore, it is important for the students to be able to identify URL phishing and malware.

Malware is malicious code that often masquerades as a part of some useful software, message, document, or data. Malware exploits the existing vulnerabilities within the IT system. Some of the malicious programs require the use of a host program to hide their tracks, e.g., Trojan horses, Spyware, Viruses and Rootkits. In contrast, others such as Worms, Automated Viruses and Zombies (Botnets) exist and propagate independently. The number and variation of malware are increasing every year. There are many mechanisms to detect malware. One of the mechanisms is hash-based malware detection. In malware research, hashing is a technique to identify a malicious file. Hashing is a mathematical process that calculates a unique fingerprint/signature for a given input. The malware is run through a hash function to produce a unique hash signature. The hash signature is then used to identify malware just like a fingerprint of digital data.

In this practicum, students learn how to identify malware and malicious URLs used for phishing attacks. To do this, you will use the VIRUSTOTAL website, which is a publicly available tool to identify malware and malicious URLs. To access the VIRUSTOTAL website, please click on the following link (<https://www.virustotal.com>).

Did you click the link? What happened?

Please note that you shouldn't trust URLs and you shouldn't directly click on them, even if you are instructed to do it. In this case, you were deceived into thinking that the provided link for VIRUSTOTAL is legitimate. However, when you clicked the URL, you were redirected to a website different from VIRUSTOTAL. In this case you were directed to the TU Delft website. However, in a phishing attack you will be directed to a malicious website that can install malware on your computer and steal your personal data.

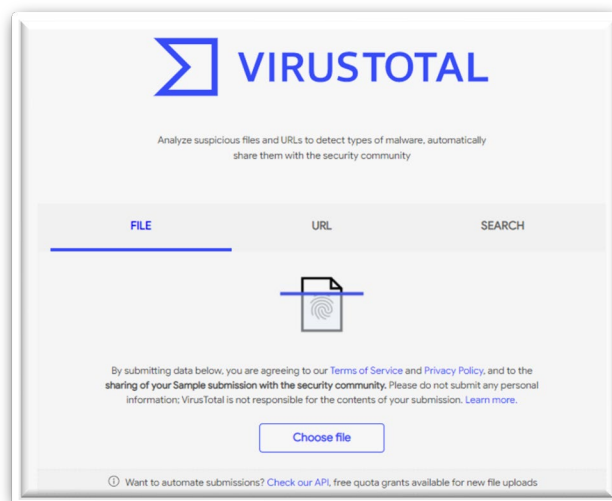
A quick way to check if a link is legitimate or not is to hover the mouse over the link, but don't click it. A text box will pop up that will display the real link you are clicking. In our case, if you hover the mouse over the VIRUSTOTAL link on page 1, the real link is www.tudelft.nl. However, for a comprehensive investigation you can use professional tools such as VIRUSTOTAL.

In this practicum, students must inspect the provided phishing links and malware files using the VIRUSTOTAL website. This tool will provide an investigation report based on the uploaded files and URLs. At the end of practicum 1, students will be able to demonstrate and explain a simple identification technique for malicious files and links. From these lab activities, students will be able to differentiate between legitimate and malicious files and links by analysing the report generated based on a public antivirus database.

Instructions:

Start by analysing suspicious files and URLs using a public scanning tool:

<https://www.virustotal.com/gui/home/url>



1. Upload each of the provided SHA256 hashes on the VIRUSTOTAL website and analyse the generated reports. Use the "Search" option in VIRUSTOTAL.
2. In the provided "Malicious URL" folder, you will find the "cybercrime-tracker" website that provides a list of malicious URLs.
 - a. Access the "<https://cybercrime-tracker.net/>" website and select 3 malicious URLs.

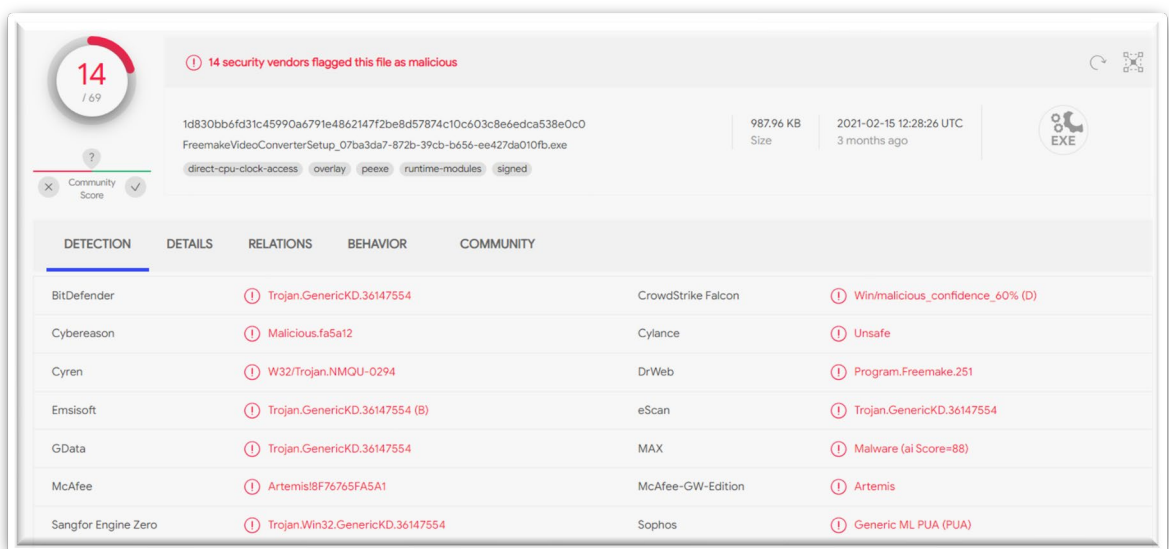
- b. Copy / paste these URLs in the VIRUSTOTAL search box under the URL tab and click search. Cybercrime tracker website also has the option to directly scan the URLs with VIRUSTOTAL by clicking the blue arrow button.



DATE	URL	IP	TYPE
23-10-2021	orepasscg.com/hrdnation/www.php	162.241.24.56	Mailer
23-10-2021	bartonmcgill.co.nz/.well-known/sand.php	172.67.204.97	Mailer

- c. Analyse the generated reports.

Here is a sample of the scanning result from VIRUSTOTAL.



14 security vendors flagged this file as malicious

1d830bb6fd31c45990a6791e4862147f2be8d57874c10c603c8e6edca538e0c0
FreemakeVideoConverterSetup_07ba3da7-872b-39cb-b656-ee427da010fb.exe

987.96 KB Size | 2021-02-15 12:28:26 UTC 3 months ago

direct-cpu-clock-access | overlay | peexe | runtime-modules | signed

DETECTION	DETAILS	RELATIONS	BEHAVIOR	COMMUNITY
BitDefender	Trojan.GenericKD.36147554	CrowdStrike Falcon	Win/malicious_confidence_60% (D)	
Cybereason	Malicious.fa5a12	Cylance	Unsafe	
Cyren	W32/Trojan.NMQU-0294	DrWeb	Program.Freemake.251	
Emsisoft	Trojan.GenericKD.36147554 (B)	eScan	Trojan.GenericKD.36147554	
GData	Trojan.GenericKD.36147554	MAX	Malware (ai Score=88)	
McAfee	Artemis!8F76765FA5A1	McAfee-GW-Edition	Artemis	
Sangfor Engine Zero	Trojan.Win32.GenericKD.36147554	Sophos	Generic ML.PUA (PUA)	

2.3 Passive Reconnaissance

Passive reconnaissance is an activity performed by adversaries to gain information about a target without actively engaging with the target's IT systems. Passive reconnaissance is often used in the Open-Source Intelligence (OSINT) activity by collecting any information from publicly available sources about the target.

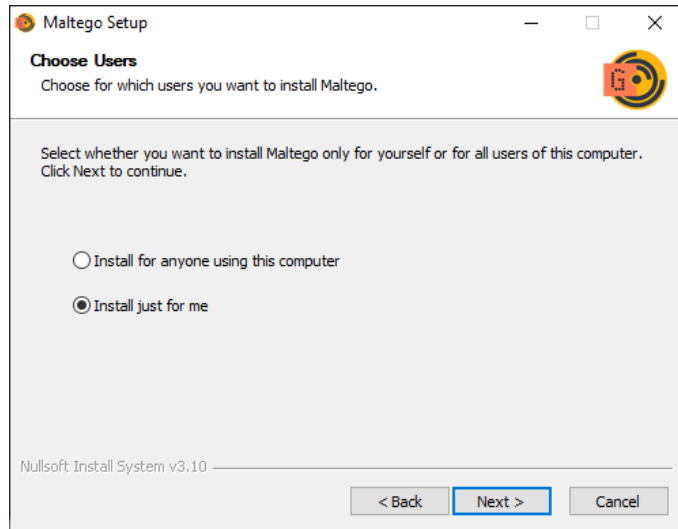
In this section, the practicum uses an OSINT tool called Maltego to gather public information about the target from the Internet. Maltego provides a graph analysis tool for gathering and connecting information about the target for a detailed investigation.

In this practicum, the students are expected to understand how OSINT works, demonstrate how to use OSINT tools, and understand how to protect data from OSINT activities.

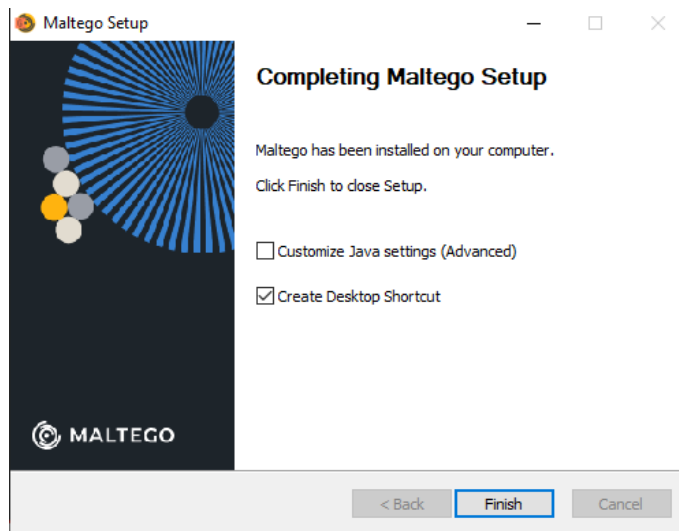
Instructions:

1. This lab is using Maltego Community Edition for free and non-commercial. Please register for a free Maltego account at this link: <https://www.maltego.com/maltego-id-registration/>
2. Install Maltego Graph Desktop through this link: <https://www.maltego.com/downloads/>. Choose the .exe option.

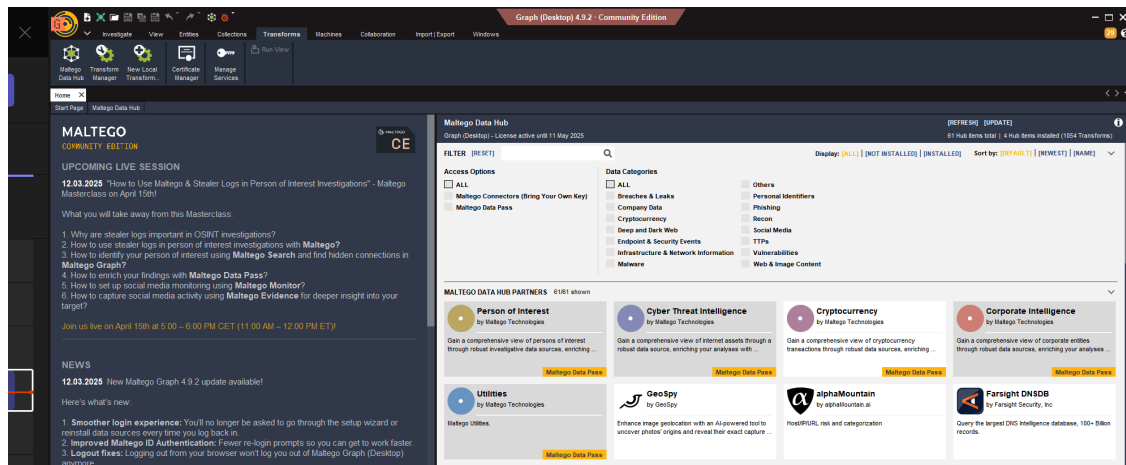
2.1 Install the Maltego Graph just for you.



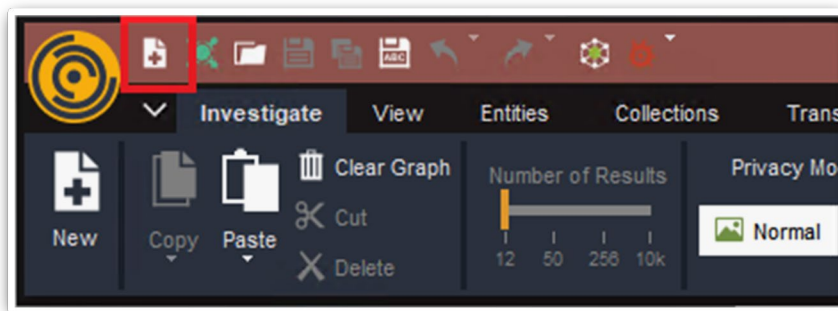
2.2 Create a Desktop shortcut.



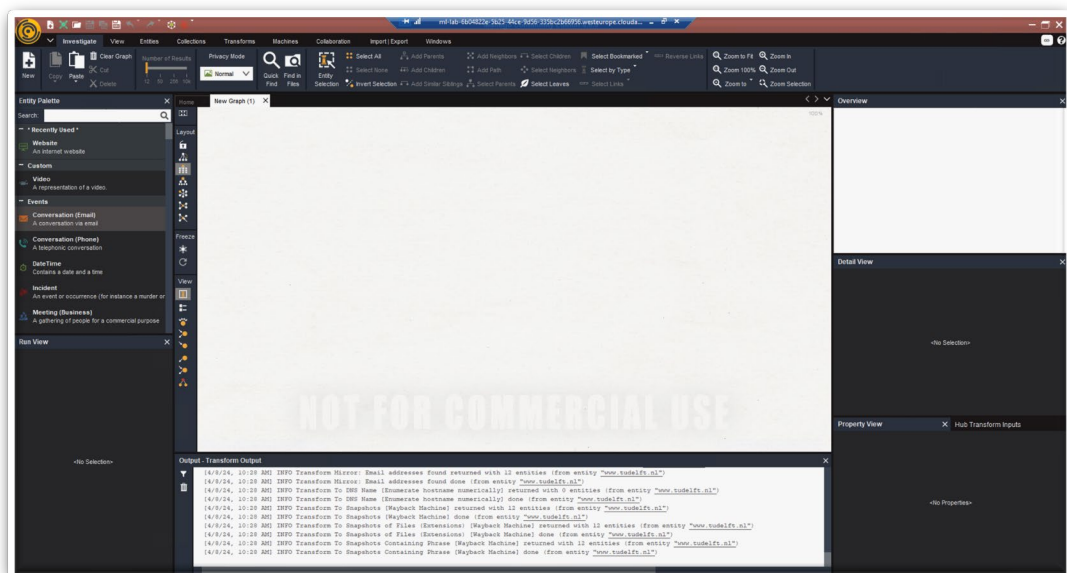
3. Open Maltego. Before you initialize your new graph, you need to install certain libraries from the Maltego community that will be used to conduct the OSINT analysis. Go to Transforms → Maltego Data Hub and install Person of Interest, Cyber Threat Intelligence, Corporate Intelligence, and Utilities as shown below.



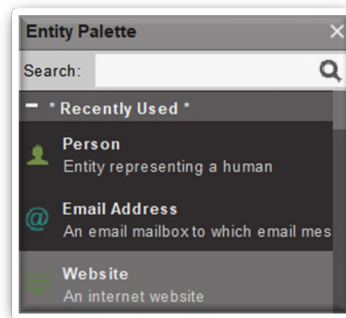
4. Go to New and create a new graph.



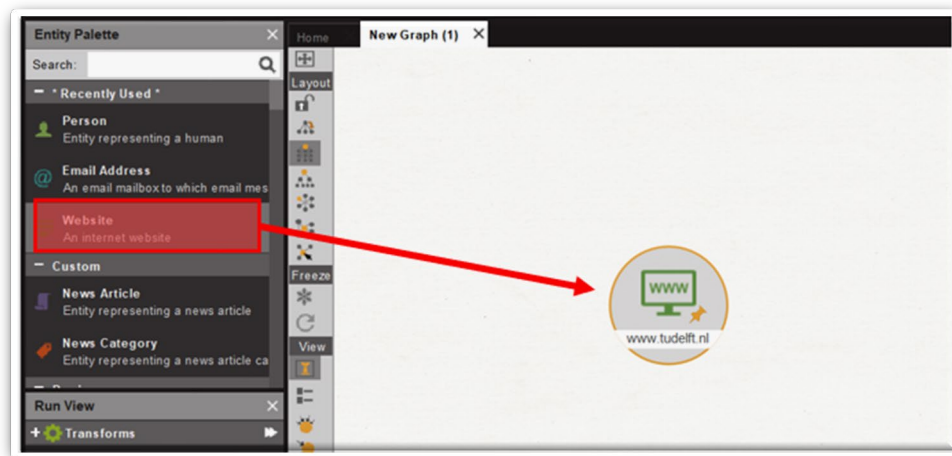
5. When you create a new graph, there is a Maltego white empty page (New Graph).



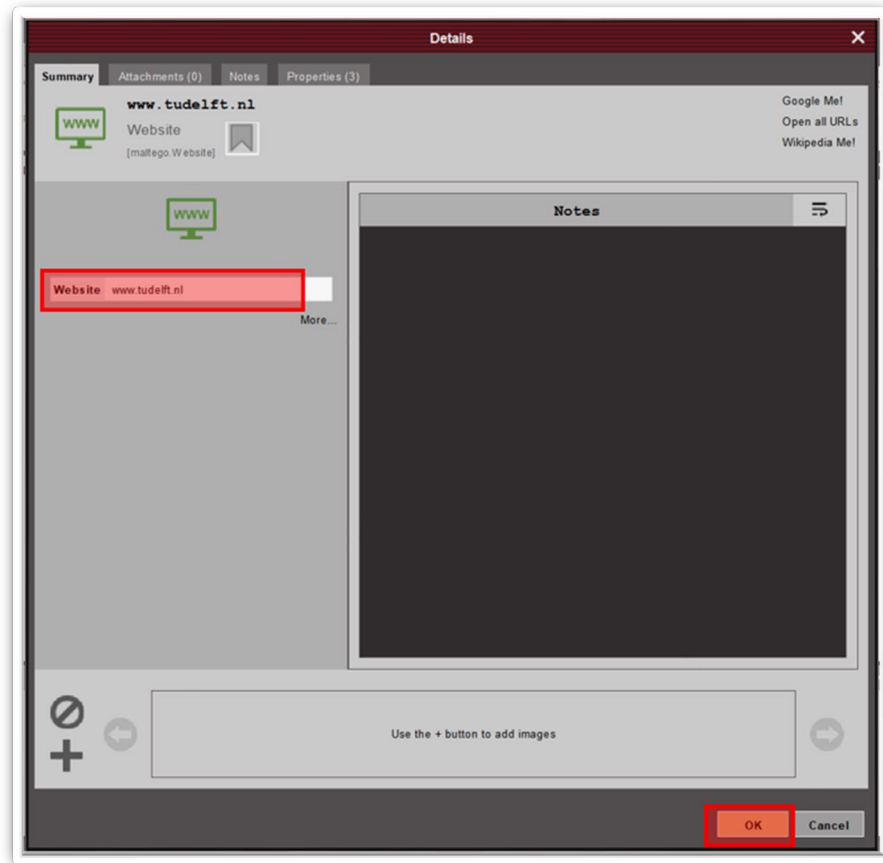
6. Choose Website from the Entity Palette.



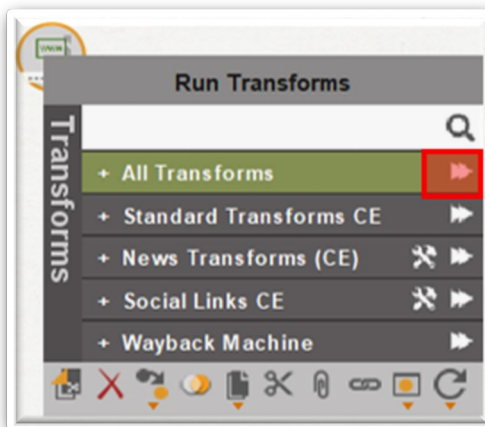
7. Drag and drop the Website element from the Entity Palette to the empty white page.



8. Double click on the Website icon on the white page, type www.tudelft.nl in the Website field, and then click OK.

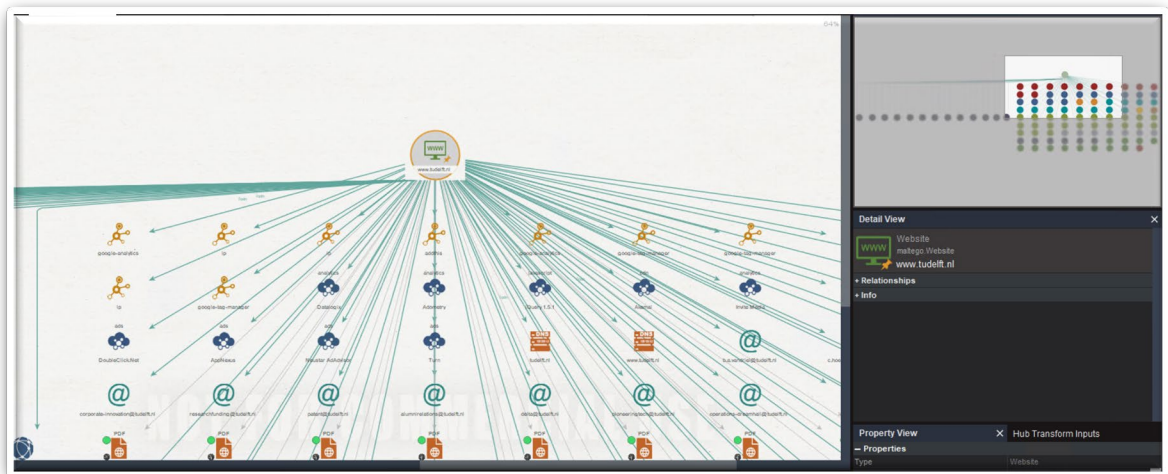


9. Back to the main page, now right-click on the Website iconrun all available transforms. Click the arrow (>>) on the right of the each individual Transform



10. After running the search, Maltego shows a graph on the main screen. You can click the icons to check more information about the target. You can also perform a further search on each icon. Right-click on the selected icon and apply specific transforms. Click the arrow

(>>) on the right of each specific transform. You can also investigate the databases that are used for each transform.



11. The following websites can also be used to find out more info about the target:

- a. Usernames search engine: <https://checkusernames.com/>
- b. Data breach database: <https://haveibeenpwned.com/>
- c. Devices search engine: <https://www.shodan.io>
- d. Google Dorks search: <https://www.techtarget.com/whatis/definition/Google-dork-query>

12. Use the instructions above to find information about yourself from the Internet. Repeat the steps above but select “Person” instead of “Website” from the “Entity Palette”.

- a. Please note that this practicum is meant strictly for educational purposes only. It is meant to make you aware that any data you share about yourself on the Internet can also be found and used by cyber criminals against you.
- b. Be careful when sharing your personal information to protect yourself from being targeted by cyber criminals.

2.4 Assignment

In this assignment, each student must write an individual report based on the activities in practicum one. During the practicum session, you are expected to collect your results by saving your files and/or taking screenshots and sending them to your email. This data will be used for your assignment and included in the practicum report. Follow the report Guidelines at the end of the Practicum Manual for detailed information about the reports.

Answer the following questions in your report, provide the practicum results using the saved data and images to explain your answers, and follow the Report Guidelines to write the report.

1. Are all the provided sample files malware or not? Explain why or why not? From the VIRUSTOTAL file scan, in the detailed report, there is information about the file signature(s). Explain how a file signature can be used to identify malware.
2. You selected and scanned 3 malicious URLs using VIRUSTOTAL. Explain your findings regarding the characteristics of the malicious URLs / websites.
3. Based on your investigation using Maltego, give an example of a case study and explain how OSINT can be used by adversaries to conduct a cyber attack or cyber crime. Use your Maltego search results to support your case study.
4. Provide your suggestions on how to prevent OSINT activities in a cyber attack or cyber crime based on your case study in question 3.

Cryptography Applications

3.1 Objectives

- Understand the application of symmetric and asymmetric cryptography.
- Demonstrate symmetric encryption and decryption using the Advanced Encryption Standard (AES).
- Demonstrate asymmetric encryption and decryption using the Rivest-Shamir-Adleman (RSA).
- Demonstrate different digital signature algorithms, including RSA Signature Scheme with Appendix – Probabilistic Signature Scheme (RSASSA-PSS, ak.a., PKCS#1 PSS) and Elliptic Curve Digital Signature Algorithm (ECDSA).

3.2 Introduction

In this practicum, students will demonstrate and evaluate the usage of different cryptography schemes using the provided Jupyter Notebook in GitLab. By making use of the already provided and implemented functions, the student will be communicating with another student using an unsecure communication channel. To simplify the exchange of messages between students, this practicum uses JSONbin.io website and its public API. Keep in mind that this is for real an unsecure communication channel given that ANY message published on the website can be read by ANYONE on the public internet.

Steps to use JSONbin.io API:

- Create an account and sign in to JSONbin.io
- From your JSONbin.io dashboard, under the "API KEYS" tab, you can obtain your personal X-Master-Key that is used to access the API. This key is unique to each JSONbin.io account and should be kept private.
- From your JSONbin.io dashboard, under the "BINS" tab, you can create a new PUBLIC bin. You can insert the following JSON to successfully initialize it: {"Lab":"Crypto"}. Note: Remember to set the bin as public.
- Save your BIN ID. This will be your "unsecure upload communication channel". Your published message will be saved in this BIN.

- Ask your friend to provide their BIN ID. This will be the "unsecure download communication channel" to be used for receiving the messages from your friend.

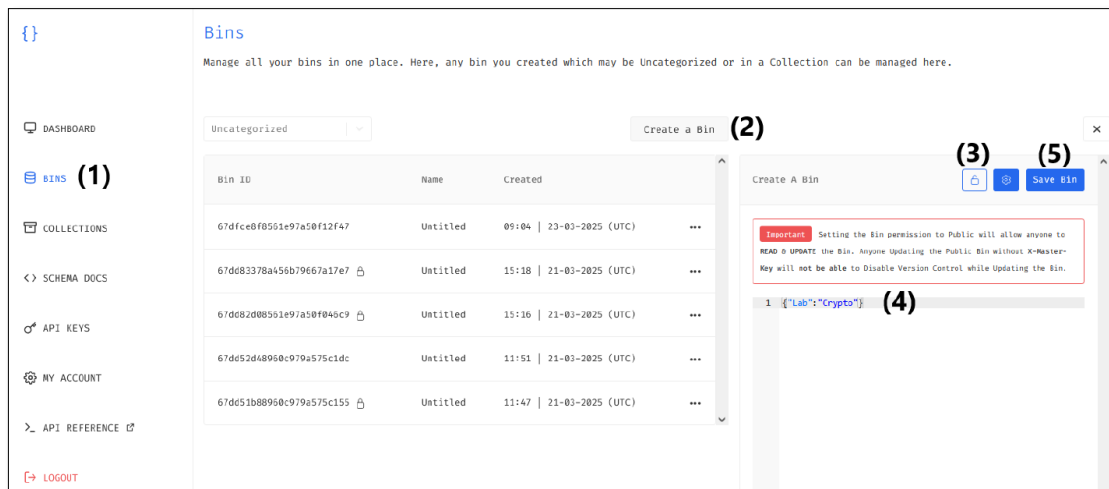


Figure 2.1: JSONbin.io step-by-step guide.

Utility Functions:

In the "Utility functions" chapter, you can find all the functions to be used to upload/download messages to JSONbin, encrypt/decrypt messages using AES/RSA, generate RSA/EC keys, and generate/validate digital signatures using PKCS#1 PSS and ECDSA. Also, examples on how to use these functions are provided. The code in the "Utility functions" chapter is not meant to be modified; it is there to help you complete the tasks in the "Assignment" chapter.

3.3 Encryption and Decryption using AES

Symmetric cryptography, also known as secret-key cryptography, is a method in which a single pre-shared key is employed for both encryption and decryption of data. This approach relies heavily on the confidentiality of the key itself, as anyone possessing the key can access and interpret the encrypted information. Due to this requirement, entities utilizing symmetric cryptography must securely exchange and manage the secret key prior to any communication. This initial exchange often presents practical challenges, as it requires a secure channel or prior trust between parties to prevent unauthorized interception.

One of the primary strengths of symmetric cryptography is its efficiency. Algorithms utilizing symmetric keys, such as the Advanced Encryption Standard (AES), are computationally faster and require fewer resources compared to their asymmetric counterparts. AES is particularly noteworthy and widely adopted globally due to its robust security characteristics. Chosen by the U.S. National Institute of Standards and Technology (NIST), AES supports various key lengths, most commonly 128, 192, or 256 bits, and modes of operations, allowing users to select a security level appropriate for their needs.

Symmetric cryptography is extensively utilized in scenarios where speed and performance are critical, such as encrypting large volumes of data or securing communication channels within

controlled environments. Common use cases include securing data at rest, such as files stored on a hard drive, or encrypting data in transit over internal network systems, like those within corporations or government institutions. However, the primary challenge remains the secure exchange of keys, which often requires additional cryptographic protocols to facilitate safe key management.

In the following, a brief description of the utility functions implemented in the Jupyter notebook and meant to be used to complete the assignment is provided. More details about the required inputs and expected outputs can be found in the Jupyter notebook.

Available functions:

- `encrypt(message, passphrase, key_length, mode) → (nonce/iv +) encrypted_message (+ tag)`
- `decrypt(ciphertext, passphrase, key_length, mode) → message`

Supported modes of operation:

- Electronic Codebook Mode (ECB)
- Cipher Block Chaining Mode (CBC)
- Cipher Feedback Mode (CFB)
- Output Feedback Mode (OFB)
- Counter Mode (CTR)
- Counter with CBC-MAC mode (CCM)
- Galois/Counter Mode (GCM)
- Offset Codebook Mode (OCB)

Supported key length:

- 128
- 192
- 256

Symmetric encryption assignment

In this part, the student is expected to experiment using AES cryptographic algorithm based on the provided function in the Jupyter notebook. The students are expected to work in pair to exchange encrypted message through JSONbin.io. The provided implementation is used to answer the assignment questions. Students will explore AES encryption by performing the following tasks:

1. Encrypt a message: Use AES with a chosen mode of operation and key length to encrypt a given message using a pre-shared secret key.
2. Publish to JSONbin: Upload the encrypted message to JSONbin.
3. Retrieve and Decrypt: Fetch an encrypted message previously published by your friend on JSONbin and decrypt it using the correct passphrase, key length, and mode of operation.
4. Analyze AES-GCM Behaviour: Encrypt and decrypt the same message multiple times using AES-GCM-128 and explain why the ciphertext varies each time.

5. Test Mode & Key Length Variations: encrypt a message using six different combinations of AES modes and key lengths, then analyze how these factors affect ciphertext length.
6. Discussion: Briefly explain the impact of mode of operation selection, key length, IV/nonce, and authentication tags on ciphertext size and security.

3.4 Encryption and Decryption using RSA

Asymmetric cryptography, also known as public-key cryptography, uses two mathematically linked keys: a public key (which is shared openly) and a private key (which is kept secret). Unlike symmetric cryptography, which relies on a single shared key for encryption and decryption, asymmetric systems enable secure communication by exchanging only the public keys among communicating entities. The public key encrypts data or verifies digital signatures, while the private key decrypts or creates signatures. This separation allows secure messaging (e.g., email encryption) and authentication (e.g., digital certificates), forming the backbone of modern internet security protocols like Transport Layer Security (TLS)/ Secure Sockets Layer (SSL).

The RSA algorithm, a pioneering asymmetric method, relies on the computational difficulty of factoring large prime numbers. To generate keys, two large primes p and q are multiplied to create a modulus $n = p \times q$. Using Euler's totient function $\phi(n) = (p-1)(q-1)$, a public exponent e is chosen such that it is coprime to $\phi(n)$. The private exponent d is computed as $d \equiv e^{-1} \bmod \phi(n)$. Encryption involves raising a message m to e and applying the modulo operation with divisor equal to n . On the contrary, decryption involves raising the encrypted message c to d , and, as before, applying the modulo operation with divisor equal to n . Security hinges on the infeasibility of factoring n to derive p and q , which would expose d . RSA remains foundational for secure data transmission, digital signatures, and key exchange in hybrid systems (e.g., combining RSA with symmetric AES for efficiency). Despite requiring longer keys (2048+ bits) to resist modern factorization attacks, its simplicity and reliability ensure its use in HTTPS, VPNs, and blockchain.

In the following, a brief description of the utility functions implemented in the Jupyter notebook and meant to be used to complete the assignment, is provided. More details about the required inputs and expected outputs can be found in the Jupyter notebook.

Available functions:

- `generate_rsa_keys(bits) → private_key, public_key, rsa_components`
- `save_rsa_keys(private_key, public_key, private_key_filename, public_key_filename) → None`
- `import_rsa_key(key_filename) → RSA key object`
- `read_rsa_key(key_string) → RSA key object`
- `encrypt_rsa(message, public_key) → encrypted_message`
- `decrypt_rsa(encrypted_message, private_key) → message`
- `sign_message_pss(message, private_key) → signature`
- `verify_signature(message, signature, public_key) → True/False`

Supported RSA key sizes (in bits):

- 1024

- 2048
- 3072
- 4096

Asymmetric encryption assignment

In this part, the student is expected to experiments using RSA cryptographic algorithm based on the provided functions in the Jupyter notebook. The students are expected to work in pair to exchange encrypted messages through JSONbin.io. The provided implementation is used to answer the assignment questions. Students will work with RSA encryption by performing the following tasks:

1. **Generate and save RSA Key Pairs:** Create a new pair of RSA private/public keys with a selected key length and save them.
2. **Print Keys:** Display the generated private and public keys in a readable format.
3. **Import RSA Keys:** Load previously saved RSA keys from files for encryption and decryption.
4. **Encrypt a Message:** Request a friend's public key, use it to encrypt a confidential message, and publish the ciphertext to JSONbin.
5. **Retrieve and Decrypt:** Fetch an encrypted message previously published by your friend on JSONbin and decrypt it using the correct private key.
6. **Experiment with Key Lengths:** Generate RSA keys with different key lengths (1024, 2048, 3072, 4096 bits) and analyze the key components (n, e, d, p, q).
7. **Discussion:** Briefly compare how key length impacts security, performance, and ciphertext size.

3.5 Elliptic Curves Digital Signature Algorithm (ECDSA)

The Elliptic Curve Digital Signature Algorithm (ECDSA) is a cryptographic protocol used to generate and verify digital signatures, leveraging the mathematical properties of elliptic curves over finite fields. As an evolution of the Digital Signature Algorithm (DSA), ECDSA replaces traditional modular arithmetic with elliptic curve operations, enabling stronger security with shorter key lengths compared to RSA or DSA. For instance, a 256-bit ECDSA key offers security comparable to a 3072-bit RSA key. This efficiency makes ECDSA ideal for resource-constrained environments like mobile devices, blockchain systems (e.g., Bitcoin and Ethereum), and IoT applications. Its adoption is widespread in protocols such as TLS, SSH, and digital identity frameworks, where both speed and compact signature sizes are critical.

ECDSA operates by generating a private key (a random integer) and a corresponding public key (a point on the curve derived by multiplying the private key with a predefined base point). To sign a message, the sender hashes the data and computes a signature using their private key and a cryptographically secure random nonce. The signature consists of two components, (r, s) , generated through elliptic curve point multiplication and modular operations. Verification involves recalculating the curve point using the sender's public key, the signature, and the message hash,

confirming it matches the original r value. Security relies on the elliptic curve discrete logarithm problem (ECDLP) with computational infeasibility of deriving the private key from the public key.

In the following, a brief description of the utility functions implemented in the jupyter-notebook and meant to be used to complete the assignment is provided. More details about the required inputs and expected outputs can be found in the jupyter-notebook.

Available functions:

- `generate_ecc_keys(curve) → private_key, public_key`
- `import_ecc_key(key_filename) → ECC key object`
- `read_ecc_key(key_string) → ECC key object`
- `export_ecc_key(key) → None`
- `sign_message_ecc(message, private_key) → signature`
- `verify_signature_ecc(message, signature, public_key) → True/False`

Signature curves:

- NIST P-192
- NIST P-224
- NIST P-256
- NIST P-384
- NIST P-521
- Ed25519
- Ed448

Digital signatures assignment

In this part, the student is expected to experiment using RSASSA-PSS and ECDSA cryptographic algorithms based on the provided function in the Jupyter notebook. The implementation is used to answer the assignment questions. The students are expected to work in pair to exchange encrypted and signed messages through JSONbin.io. Students will explore digital signatures using RSASSA-PSS and ECDSA. They will complete the following tasks:

1. Sign and Encrypt a Message: Encrypt a message using RSA, generate a digital signature, and publish both to JSONbin.
2. Retrieve and Verify: Fetch an encrypted message and its signature from JSONbin, verify its authenticity using the sender's public key, and decrypt it.
3. Key Usage Discussion: Explain which keys were used for encryption/decryption versus signature generation/validation.
4. Signature Experiments: Generate and compare signatures using different key types (e.g., RSA 1024-bit and 4096-bit, ECDSA NIST P-192, P-224, and P-521).
5. Key & Signature Analysis: Briefly discuss how key type and length affect signature size, security, and performance in RSA vs. ECC.

3.6 Assignment

In this assignment, each student must write an individual report based on the Jupyter notebook implementation.

Answer the following questions in your report, provide the practicum results using the saved data and images to explain your answers, and follow the Report Guidelines to write the report.

1. Symmetric encryption:
 - a. Describe the various steps you followed to encrypt, upload, download, and decrypt the secret messages.
 - b. Discuss which information you needed to share with your friend to enable the exchange of the encrypted messages.
 - c. Briefly discuss how different modes of operation and key lengths affect ciphertext length and security.
2. Asymmetric encryption:
 - a. Describe the various steps you followed to encrypt, upload, download, and decrypt the secret messages.
 - b. Discuss which information you needed to share with your friend to enable the exchange of the encrypted messages.
 - c. Regarding the previous point, compare with what was required when using symmetric encryption. What are the pros and cons of each cryptography scheme?
 - d. How does the RSA key length relate to the length (in bits) of the RSA components (i.e., n, p, q, e, d)?
3. Digital signatures:
 - a. Describe the various steps you followed to encrypt, sign, upload, download, decrypt, and verify the secret signed messages. In particular, highlight which private and public keys were used for each operation.
 - b. Briefly discuss and compare (qualitatively) how PKCS#1 PSS and ECDSA differ in terms of: private and public key length, signature size, security, and performance

Cyber-Physical Data Correlation

4.1 Objectives

- Understand the nature of cyber-physical data of power grids.
- Apply statistical techniques to differentiate between faults and spoofing cyber attacks in digital substations.
- Evaluate and analyze the performance of the statistical techniques.

4.2 Anomaly Detection and Correlation of Power Grid Cyber-Physical Data

Anomaly detection and correlation between cyber-physical data in power systems is complicated due to the inherent differences of the data sources, i.e., physical data is continuous and time-varying, while cyber data is typically discrete and event-driven. Thus, correlating between them to identify anomalies is non-trivial.

In this lab exercise, students will apply the techniques taught in class towards correlation of cyber-physical data from digital substations. Students will apply statistical techniques to differentiate between faults and spoofing cyber attacks in digital substations.

4.3 Assignment

1. Familiarization and data visualization.
Load the provided CPS dataset and explore it. Try to answer/investigate the following:
 - a. What types of signals are there? What do they represent?
 - b. Are the signals correlated? Do they show cyclic/periodic behaviour?
2. ARMA model
Try to implement an ARMA model and use it to predict future samples for both physical and cyber data. What is the output performance and why?
3. SARMA model
Repeat the above with a Seasonal – ARMA (SARMA) now instead. Do you notice any significant changes in output performance? Try to justify/reason why this could be.

4.4 References

- Abedi, V. S. Rajkumar, A. Ştefanov and P. Palensky, "Towards Real-Time Distinction of Power System Faults and Cyber Attacks," 2023 IEEE Power & Energy Society General Meeting (PESGM), Orlando, FL, USA, 2023, pp. 1-5. [\[link\]](#)

Operational Technology Traffic Analysis

5.1 Objectives

- Understand the characteristics of Operational Technology (OT) communication traffic.
- Implement OT traffic time-series analysis in Python.
- Apply traffic dispersion graph for OT traffic analysis.

5.2 Operational Technology Networks

The modern power grid is a complex system that relies on information and communication technologies for monitoring and controlling in real-time the physical processes in power plants and substations from control centres. The OT communication networks are used for power system operation. They ensure the availability, stability and safety of critical power system processes. Information Technology (IT) communication networks are also used in the offices for various managerial and administrative functions such as network planning, scheduling, and asset management. IT communication systems are NOT used for power system operation.

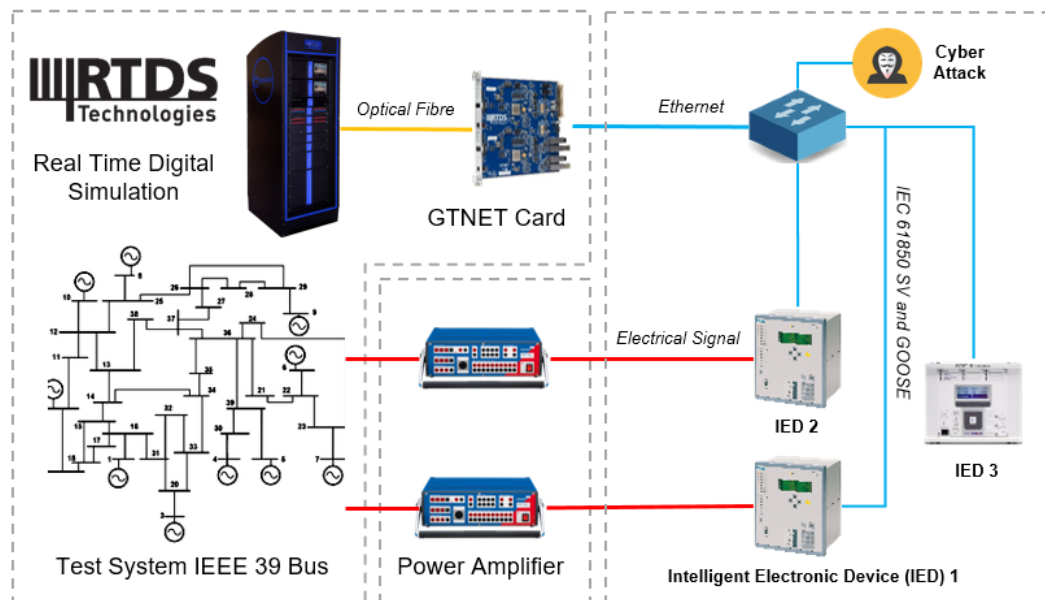


Figure 4.1 Hardware in the loop simulation of OT communication traffic with RTDS

The IT and OT communication network traffic characteristics exhibit notable distinctions. The network traffic in OT systems originates from automated processes with deterministic and

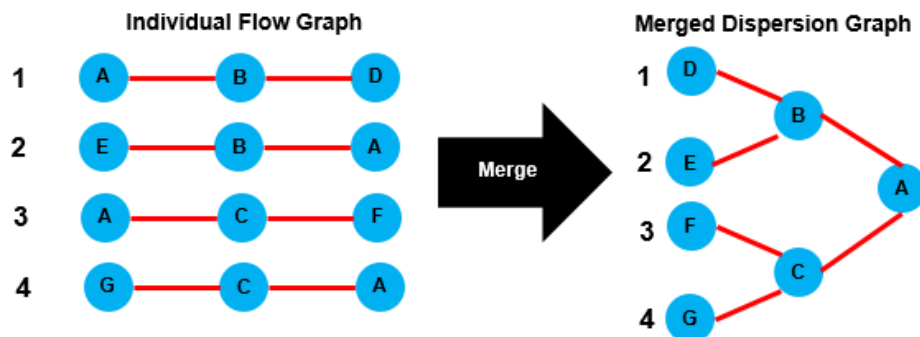
homogeneous behavior. In contrast, the traffic in IT systems primarily comprises of user-generated data that has a stochastic behavior. In this practicum, students learn how to analyze the OT traffic from hardware-in-the-loop simulations of the Real-Time Digital Simulator (RTDS) and protective relays depicted in the Figure 4.1. The analysis is performed based on the captured traffic from the network switches saved as CSV files. The students conduct the traffic analysis using standard Python libraries, i.e., pandas and matplotlib.

5.3 Traffic Dispersion Graph

Traffic Dispersion Graph (TDG) is an analytical model for communication traffic monitoring and analysis. The core idea for TDG is derived from the behavior of hosts in a communication network. Therefore, the flow of OT network traffic can be analyzed based on the interactions between all hosts in the communication network. Based on this analysis, information related to communication sources and destinations is extracted. Furthermore, TDG represents nodal information using graph structures. Every host in a network is represented by a single node in a graph. On the other hand, communication between hosts is represented by connectivity between nodes, i.e., graph edges. The figure below shows the TDG generation processes. Firstly, information on the IP or MAC address source and destination from flowing packets in the network is collected in the information table. Information about the path between two addresses is added based on prior knowledge of the network topology. The information in the table is then used to create an individual flow graph. Finally, all individual flow graphs are merged into a dispersion graph which provides an overall topology of the network. In this practicum, the students learn how to generate TDG from the OT traffic. The TDG is generated based on the CSV files using the 'networkx' library in Python.

Collected Information Table

No	IP source	IP destination	Path
1	A	D	B
2	E	A	B
3	A	F	C
4	G	A	C



5.4 Assignment

Part 1: Network Traffic Time-Series Analysis

1. Plot the traffic data from the provided CSV files.
 - a. Visualize and analyze all the protocols present in the dataset. (Hint: try using *scatter plots* or *heatmaps*, *boxplots* or *clustering*).
 - b. Limit the visualization to type GOOSE. Is the signal cyclic? Explain why or why not?
2. Perform resampling of the times-series data to merge the length of every x milliseconds and plot the result.
 - a. Plot different variations of resampling time in milliseconds, for example, $x = \{100, 200, 500, 1000, 2000\}$ etc.
 - b. Show the plot for the best resampling time for all protocols.
 - c. Show the plot for the best resampling time for GOOSE only.
3. Based on the plot results and CSV files, compare and comment on the characteristics and differences between IT and OT protocols.
4. Analyze and compare the traffic from all CSV files. Which file is most likely to contain anomalous traffic? Support your answers with representative plots.

Part 2: Traffic Dispersion Graph

1. Plot graphs from the CSV files using Networkx. Combination of sources and destinations in the CSV files serve as edge lists for the Networkx graph.
2. Modify the graph by adding switch nodes, such that all traffic flows via the switch node.
3. How many total nodes/devices exist for each graph?
4. Briefly identify and explain the node characteristics for each graph.

5.5 References

- A. Presekal, A. Ştefanov, V. S. Rajkumar and P. Palensky, "Attack Graph Model for Cyber-Physical Power Systems Using Hybrid Deep Learning," in IEEE Transactions on Smart Grid, vol. 14, no. 5, pp. 4007-4020, Sept. 2023. [[link](#)]
- A. Presekal, A. Ştefanov, V. S. Rajkumar and P. Palensky, "Cyber Forensic Analysis for Operational Technology Using Graph-Based Deep Learning," 2023 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm), Glasgow, United Kingdom, 2023, pp. 1-7. [[link](#)]

Anomaly Detection and Time Series Classification

6.1 Objectives

- Analyze anomaly detection for power grid OT network traffic.
- Apply traffic dispersion graph for OT traffic analysis with OT communication traffic throughput for anomaly detection.
- Apply a time-series classification machine learning model for anomaly detection in power grid OT traffic.

6.2 Cyber-Physical Power System Co-Simulation

Comprehensive Cyber-Physical System (CPS) models are needed for research on cyber security of power grids. They are used to simulate the power systems along with their associated IT-OT communication networks and cyber events. This practicum uses co-simulation of the power grid and IT-OT systems, as depicted in Figure 5.1. The CPS model provides time-domain measurement data from substation bays, e.g., buses, lines, and generators, in the form of active and reactive power, voltage, and current measurements. All measurement data is then delivered from the substation to the control center via a WAN as SCADA telemetry. The SCADA data is also stored in local databases located in substations and the control center. For the cyber system, every node in the OT network is emulated using operating system-level virtualization. The network connectivity between substations, WAN, and control center is realized through network virtualization and SDN. With this configuration, the developed CPS architecture can model and simulate realistic OT network traffic for the power system.

The OT network is modeled based on custom functions for every device in the communication network. The measurement devices represent components, such as Merging Units (MUs), Remote Terminal Units (RTUs), and Intelligent Electronic Devices (IEDs). These devices perform data acquisition from the power grid, with a SCADA sampling rate of one sample per second. Legitimate control commands from the control center modify the set points for power grid controllers in real-time. For example, a control command can set a circuit breaker to open or close, set values for voltage, and active power set points of generator automatic voltage regulators and governors. The measurement values and control set points are communicated across the OT network using Transmission Control Protocol/Internet Protocol (TCP/IP) packets.

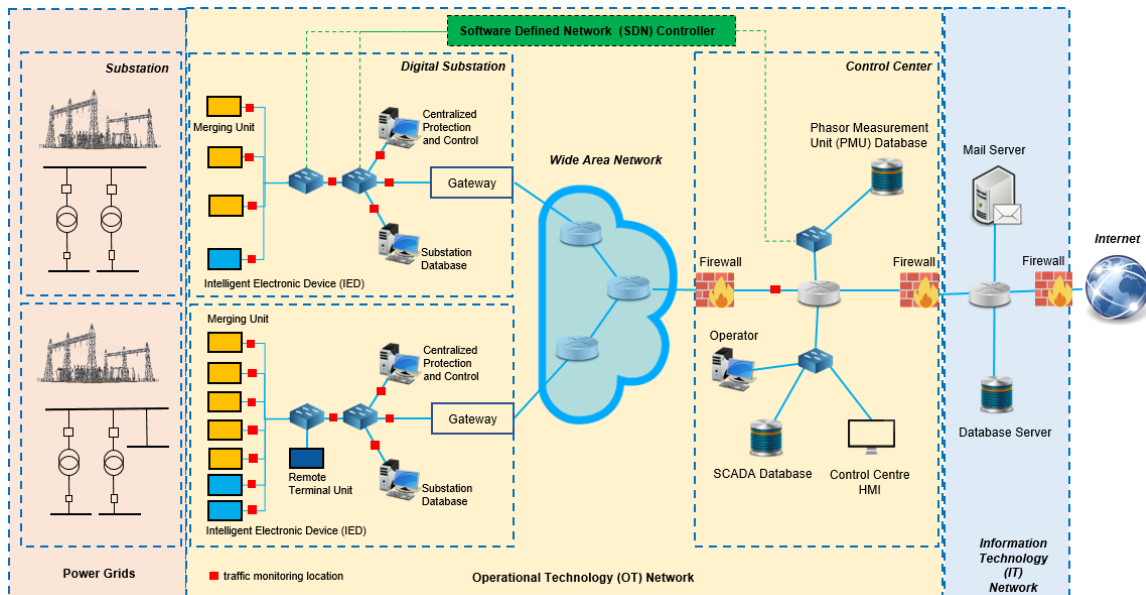


Figure 5.1. Co-simulation of power grid and IT-OT systems

In this practicum, traffic captures from a substation are provided as CSV file. Each device in the OT network is assigned with a unique IP addresses of 100.x.0.x. The network topology of the substation is depicted in Figure 5.2 and contains 11 nodes from A to K. The nodes A to J are located inside the substation and node K represents a gateway that bridges the connection between the substation and the Wide Area Network (WAN) and control center.

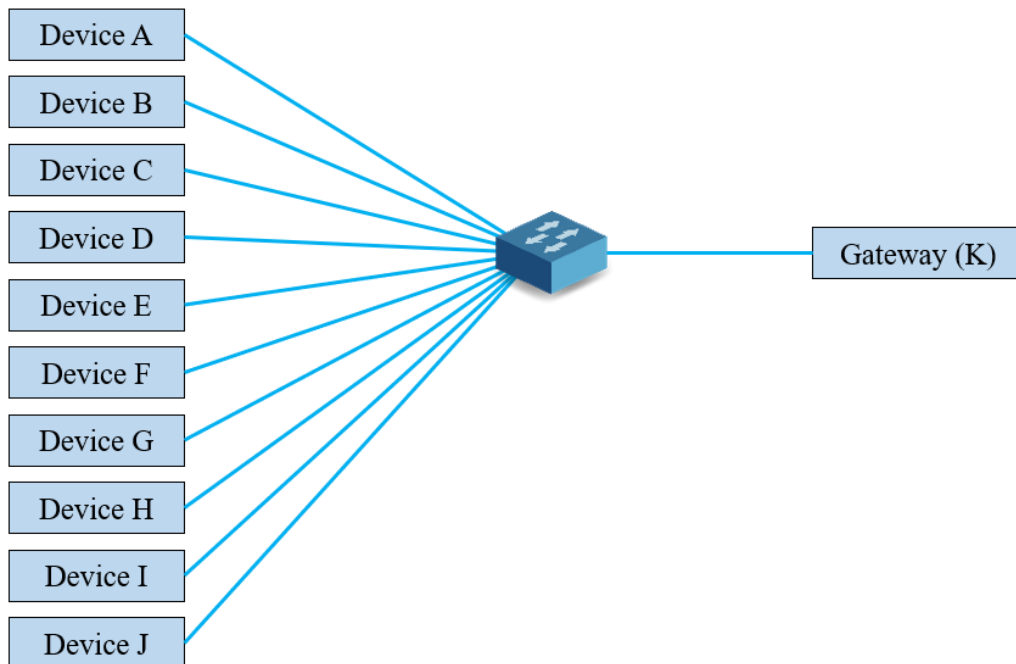


Figure 5.2. OT communication network topology of the substation

Based on the provided CSV datasets, students are expected to perform anomaly detection. Three different scenarios are provided, i.e., scenario X, Y and Z. Each scenario represents different conditions of OT traffic in the substation, i.e., cyber attack and normal operation. For traffic throughput and TDG analysis, this practicum uses pcap_X.csv, pcap_Y.csv, and pcap_Z.csv.

6.3 Anomaly Detection and Time Series Classification

Time Series Classification (TSC) has been widely implemented for anomaly detection in the literature. TSC uses sliding windows to classify sequential data into different categories or classes, where the input data is represented as a series of observations recorded over time. In this approach, the time series data is partitioned into smaller segments through the utilization of a sliding window mechanism, followed by the application of classification algorithms to these segments. Figure 5.3 shows time series data of OT traffic with sliding windows. Based on the pattern of the traffic, the algorithm will classify the traffic into a normal or anomalous.

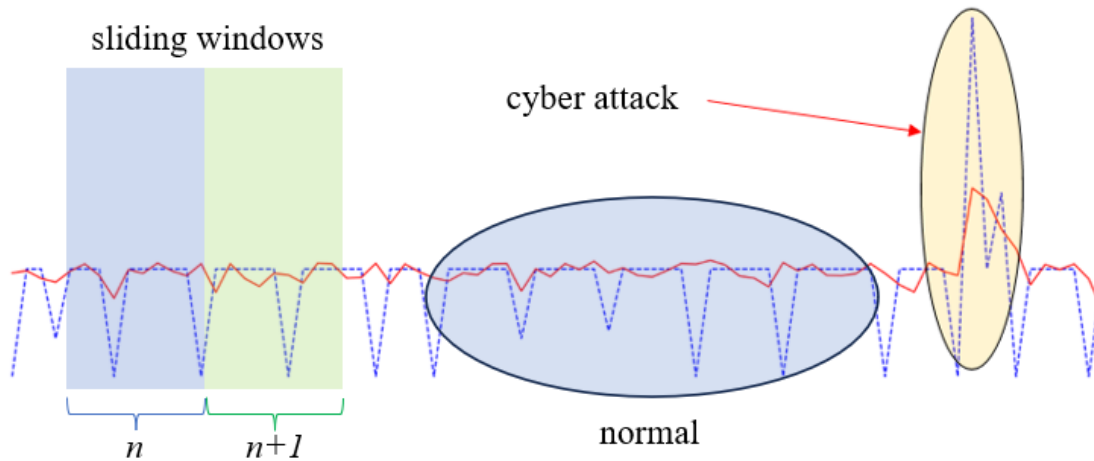


Figure 5.3. Time series data of OT traffic with sliding windows

In this practicum, students use a TSC machine learning model to detect anomalies in the OT communication network traffic throughput. This practicum uses the training dataset *tsc_training_data.csv* and test dataset, *test_X.csv*, *test_Y.csv*, and *test_Z.csv*. Initially, students train a machine learning model using the training dataset. The trained model is then used to classify the time series test data X, Y, and Z.

6.4 Assignment

1. Network Traffic Analysis Using Throughput and TDG.

Amongst the 3 scenarios (pcap_X.csv, pcap_Y.csv, and pcap_Z.csv):

- Which of them are classified as normal traffic and cyber attack traffic? Support your answer with data analysis and plots.
- Identify what are the adversary IP address and attack target IP addresses for each scenario.

2. *Time Series Classification using Machine Learning.*

Use the labelled dataset *tsc_training_data.csv* to train a TSC machine learning model.

Note: You are allowed to test and modify the model and hyperparameters to improve classification performance!

Implement the trained machine learning model to classify unlabelled time series data of *test_X.csv*, *test_Y.csv*, and *test_Z.csv*. Based on the prediction results, answer the following questions:

- a. Among all test data, which of them are classified as normal traffic and cyber attack traffic?
 - b. Identify the impact of cyber attack on the time series traffic. Support your answer with data analysis and plots.
3. Provide a comparative analysis of the two methods (TDG and TSC) used in Parts 1 and 2. List the advantages and drawbacks for each method.
 4. Based on the lectures and assignment, provide suggestions/ideas on how to detect anomalous traffic in power grid operational technology networks?

6.5 References

- A. Presekal, A. Ştefanov, V. S. Rajkumar and P. Palensky, "Attack Graph Model for Cyber-Physical Power Systems Using Hybrid Deep Learning," in *IEEE Transactions on Smart Grid*, vol. 14, no. 5, pp. 4007-4020, Sept. 2023. [[link](#)]
- A. Presekal, A. Ştefanov, V. S. Rajkumar and P. Palensky, "Cyber Forensic Analysis for Operational Technology Using Graph-Based Deep Learning," 2023 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm), Glasgow, United Kingdom, 2023, pp. 1-7. [[link](#)]

7.1 Report Requirements

1. For each lab session, you must complete the assignment(s) and write an individual report. The reports must be submitted on Brightspace, which will be checked automatically for plagiarism. We hereby remind you of the TU Delft plagiarism advice:
<https://www.tudelft.nl/en/student/legal-position/fraud-plagiarism/>
2. The deadline for the report submission on Brightspace is one week after the lab session. All assignments and reports are mandatory.
3. You make your own report for the assignment(s) and don't collaborate with other students on any questions or potential answers.
4. Please write your assignment report in English.
5. You must submit the reports before the submission deadlines. Reports submitted after the deadline will be graded with a 0. The deadline for the report submission is one week after the lab session. Please check Brightspace for more details.
6. Each student must submit their original reports in PDF file format with a **maximum of five pages (refer the table in Section 1. Lab Assignments)** using the A4 size, excluding the cover page and reference page. The report text must be written with Times New Roman font size 11.
7. For Lab Session 2, 3, 4 and 5 the link of Jupyter Notebook report must be included on top of the first page of the PDF report (not on the cover page). Make sure the link is correct and accessible before submitting the PDF report on the Brightspace.
8. The title page of PDF and Jupyter Notebook should contain:
 - a. EE4410 Cyber Security of Power Grids
 - b. Assignment / lab session number and title
 - c. Student first and last name and student number
 - d. Date of submission
9. Page numbers is a must-have in the PDF report. Figures and tables need a title and a number.

10. All references used in the report must be cited properly. A list of the references must be located on the last page. The reference style is the IEEE format. Guidelines on how to cite: <https://tulib.tudelft.nl/writing-publishing/how-to-cite/>

7.2 Report Scoring Matrix

Grade	Quality of figures and tables	Quality of text and analysis
0	IF IN ALL FIGURES/TABLES: Labels/captions/legend missing. Illegible text. Overcrowding. Unidentifiable elements (e.g., curves). Poor resolution.	Incomplete description of the concept and results. No reflections. Document contains serious spelling and grammatical errors. Report with more than one page over-length.
1	IF IN AT LEAST TWO FIGURES/TABLES: Labels/captions/legend missing. Illegible text. Overcrowding. Unidentifiable elements (e.g., curves). Poor resolution.	Follows the indicated structure, fair descriptions of the concept and results. No reflections. Document contains some spelling and grammatical errors. Report with more than one page over-length.
2	IF IN AT LEAST ONE FIGURE/TABLE: Labels/captions/legend missing. Illegible text. Overcrowding. Unidentifiable elements (e.g., curves). Poor resolution.	Follows the indicated structure, proper description of concepts and results. Poor reflections on results. The document contains few spelling and grammatical errors. Report with one page over-length.
3	Good quality figures/tables. However, information is not properly processed. Figures and tables have unnecessary info.	Follows the indicated structure, good description of concepts and results. Limited reflections on results. Document contains only minor spelling and grammatical errors. Report with one page over-length.
4	Very good quality figures/tables. However, information is not properly processed. Figures and tables contain essential info.	Very good in terms of structure, contents, clarity, analysis, and reflections. Concepts from the lecture are understood and used to analyse the results. Document is without any spelling and grammatical errors. Report with the correct length.
5	Excellent quality figures/tables. Excellent processing and presentation of results. Only essential information is provided to convey the main findings properly.	Excellent in terms of structure, contents, clarity, analysis, and reflections. Concepts from the lecture are understood and used to analyse the results. Spelling and grammatically error-free. Report with the correct length.

LAB ACCESS

Access Guidelines

For practicum 1: Please install the following software on your **personal computer / laptop**. If possible, please bring your personal laptop also during the lab session, as it is not allowed to install the Maltego software on TU Delft Devices.

Download and install the Maltego community edition (free). Register to the website to get the free license. Link: <https://www.maltego.com/downloads/>

For Practicums 2, 3, 4, and 5: students will use Jupyter Notebooks. The reference code and datasets are provided in the following TU Delft GitLab repository:

The link will be provided before Lab Session 2.

Please note: students must log in to the [TU Delft GitLab](#) with their netID to access the repository!